Section 2.6 Related Rates

Problem Solving Procedure.

1. Draw a picture and name the variables and constants.
2. Write down the numerical information.
3. Write down what you are asked to find.
4. Write an equation that relates to the variables.
5. Differentiate.
6. Evaluate.

1. A hot air balloon rising straight up from a level field is tracked by a range finder 150 m from the liftoff point. At the moment the range finder's elevation angle is $\pi/4$, the angle is increasing at the rate of 0.14 rad/min. How fast is the balloon rising at that moment?

2. Water runs into a conical tank at the rate of $9\,\text{ft}^3/\text{min}$. The tank stands point down and has a height of $10\,\text{ft}$ and a base radius of $5\,\text{ft}$. How fast is the water level rising when the water is $6\,\text{ft}$ deep?

3. There is a rope running through a pulley and bearing a weight at one end. The other end is held 5 ft above the ground in the hand of a worker. Suppose that the pulley is 25 ft above the ground, the rope is 45 ft long, and the worker is walking away from the weight at a rate of 4 ft/s. How fast is the weight raised when the worker's hand is 21 ft away from the weight's starting position?

4. Sand falls from an overhead bin and accumulates in a conical pile with a radius that is always three times its height. Suppose the height of the pile increases at a rate of 2 cm/s when the pile is 12 cm high. At what rate is sand leaving the bin at that instant?